Project Title
(Title: Student Performance & Dropout Prediction System)

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING (25SC2107E)

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ABSTRACT

Title: Student Performance & Dropout Prediction System

The Student Performance & Dropout Prediction System is a machine learning-based application designed to predict students' academic performance and identify students who may be at risk of poor performance or dropping out. The system analyzes important factors such as attendance, internal marks, study hours, previous academic performance, and other relevant student attributes to identify patterns associated with academic outcomes.

The proposed system applies machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to build and compare predictive models. K-Means clustering and Principal Component Analysis (PCA) can also be used to identify student groups and visualize their academic patterns. The models are evaluated using appropriate metrics such as accuracy, precision, recall, F1-score, and confusion matrix. A simple Streamlit-based interface can be developed to provide predictions and visualize important student information.

The system aims to help educational institutions identify at-risk students at an early stage and provide timely academic support. Overall, the project demonstrates the practical application of data preprocessing, exploratory data analysis, supervised learning, unsupervised learning, model evaluation, and machine learning deployment in the education domain.

Signature of the Guide